

SIH INTERNAL HACKATHON: TOP 10 PROBLEM STATEMENTS

Curated Software-Only Smart Automation Solutions for College Selection Round

01 AI-Powered Mobile Grain Quality Grading & Fair Price Estimation

Tags: AgriTech • Farmer Welfare

Target: Agriculture, Mandis, and Farmer Welfare | **Feasibility:** 8.5/10 | **Impact:** 9.5/10 | **Winning Edge:** On-device YOLOv8

Aligned Govt Schemes: PM-KISAN, eNAM, Digital India

■ The Problem

At agricultural mandis, grain quality checking is done manually by traders who visually estimate impurities. Traders exaggerate defects (chaff, broken grains, discoloration) to buy crops at under-rates, systematically exploiting farmers who have no scientific verification tool.

■ Proposed Solution

A mobile app where the farmer spreads grain on a dark sheet, snaps a photo. AI performs instance segmentation to count and classify healthy grains, broken grains, discolored grains, and dust/stones. Cross-references with government MSP and live market rates to output a fair price per quintal. Works fully offline.

02 Deep-Learning Counterfeit Medicine Packaging Detection Engine

Tags: Public Health • Pharma

Target: Healthcare, Consumer Protection, Pharma | **Feasibility:** 8.0/10 | **Impact:** 9.5/10 | **Winning Edge:** Siamese Network Forensics

Aligned Govt Schemes: Ayushman Bharat, Jan Aushadhi, Digital Health Mission

■ The Problem

Counterfeit medicines plague Indian pharmacies. Fake packaging looks identical to the naked eye, leading to treatment failures and preventable deaths. Normal consumers cannot distinguish genuine from counterfeit at the pharmacy counter.

■ Proposed Solution

A verification app scans medicine strip packaging using a smartphone camera. AI analyzes micro-printing defects, font kerning shifts, logo misalignment, and packaging texture profiles that differ when counterfeiters duplicate original printing plates. Cross-references batch numbers against a secure registry.

03 Crowdsourced Pothole Detection & Municipal Work-Order Automation

Tags: Road Infrastructure • Smart Cities

Target: Road Transport, Municipal Corporations | **Feasibility:** 8.5/10 | **Impact:** 9.0/10 | **Winning Edge:** DBSCAN + Accelerometer

Aligned Govt Schemes: Smart Cities Mission, Bharat NCAP, PMGSY

■ The Problem

Indian roads are notorious for potholes causing thousands of fatal accidents annually. Municipalities rely on slow manual inspections or reactive citizen complaints, leading to massive delays. There's no systematic, real-time way to detect and track road damage.

■ Proposed Solution

A background SDK running in driver/delivery apps. When the phone is dashboard-mounted, it uses the camera + accelerometer to auto-detect potholes and geo-tag them. Thousands of reports from different vehicles are clustered into confirmed locations on a live GIS map, automatically generating municipal repair tickets.

04 Satellite-AI Stubble Burning Logistics Marketplace & Smart Allocation Engine

Tags: Environment • Agriculture

Target: Environment, Agriculture, Energy | **Feasibility:** 7.5/10 | **Impact:** 9.5/10 | **Winning Edge:** Satellite + VRP Routing

Aligned Govt Schemes: National Clean Air Programme, GOBAR-Dhan, SATAT Scheme

■ The Problem

Crop residue burning (parali) in North India causes deadly winter smog in Delhi-NCR. Farmers burn stubble because manual removal is expensive and they lack coordination to sell it to biomass/biogas companies. There's no logistics platform connecting supply (farmers) to demand (industry).

■ Proposed Solution

An automated supply-chain marketplace. Satellite remote sensing predicts harvest dates and estimates stubble volume on fields. The platform coordinates local aggregators (baler operators) to collect, pack, and transport biomass to buying plants using optimized vehicle routing within the short 30-day harvest window.

05 Autonomous Ambulance Traffic Optimization & Green Corridor API

Tags: Emergency Services • Smart Cities

Target: Urban Traffic, Emergency Medicine | **Feasibility:** 7.0/10 | **Impact:** 9.0/10 | **Winning Edge:** RL Signal Control

Aligned Govt Schemes: Smart Cities Mission, National Health Mission, 108 Ambulance Network

■ The Problem

Traffic congestion in Indian cities causes severe ambulance delays, often leading to patient deaths during transit. Traffic police manually clearing roads is slow and cannot coordinate across multiple junctions simultaneously. Every minute counts.

■ Proposed Solution

Software that integrates ambulance GPS with city Intelligent Traffic Management Systems. It calculates the ambulance's path, dynamically adjusts traffic light timings (signal preemption) along the route to create an automated "green corridor" — turning signals green before the ambulance arrives and restoring normal cycles after it passes.

06 Municipal Water Distribution Leakage & Theft Analytics Dashboard

Tags: Utilities • Jal Shakti

Target: Water Resources, Municipal Utilities | **Feasibility:** 8.0/10 | **Impact:** 8.5/10 | **Winning Edge:** Physics-Informed ML

Aligned Govt Schemes: Jal Jeevan Mission, AMRUT 2.0, Smart Cities Mission

■ The Problem

Up to 40% of piped water in Indian municipalities is lost to pipe leakage and illegal tapping before reaching households. This causes severe drinking water shortages. Municipalities cannot locate leaks without expensive, slow physical excavation.

■ Proposed Solution

A diagnostic software platform ingesting flow rate and pressure telemetry from existing bulk meters at key distribution nodes. AI detects abnormal pressure drop signatures and pinpoints leak/theft coordinates by modeling hydraulic water head loss using physics-informed machine learning, without digging up streets.

07 Multilingual Panchayat Grievance AI Router & Voice-To-Ticket Engine

Tags: Governance • Digital India

Target: Rural Governance, Citizen Grievances | **Feasibility:** 9.0/10 | **Impact:** 9.0/10 | **Winning Edge:** Bhashini + LLM Agent

Aligned Govt Schemes: Digital India, National Translation Mission, e-Governance

■ The Problem

Rural citizens cannot file formal grievances because they cannot write in English/Hindi. Paper complaints get lost in bureaucracy with no tracking. Nobody follows up. Language barriers and illiteracy exclude millions from the governance system.

■ Proposed Solution

A voice-first complaint portal. Citizens speak their complaint in any regional language or upload a photo of a handwritten letter. AI transcribes, translates, categorizes the issue, identifies the correct department, auto-generates a formal ticket, and tracks resolution SLA. Updates are sent back to the citizen via WhatsApp in their language.

08 NEP-Aligned Multilingual Adaptive Rural Tutor

Tags: Education • NEP 2020

Target: Education, Rural Literacy | **Feasibility:** 8.5/10 | **Impact:** 8.5/10 | **Winning Edge:** BKT + Offline PWA

Aligned Govt Schemes: NEP 2020, PM eVidya, DIKSHA, Digital India

■ The Problem

Rural government schools face terrible teacher-to-student ratios. Children drop out because they can't keep up. Existing EdTech (Byju's etc.) is expensive, needs constant internet, and is English-dominated — excluding rural India entirely.

■ Proposed Solution

An offline-first adaptive learning PWA acting as a personalized tutor. It continuously evaluates learning gaps, adjusts difficulty dynamically, and features an interactive voice assistant explaining concepts in the student's native language. 100% functional offline once content is cached.

09 GeM Bid Cartel & Price Manipulation Detection Engine

Tags: Procurement • Anti-Corruption

Target: Public Procurement, Government e-Marketplace | **Feasibility:** 8.0/10 | **Impact:** 9.0/10 | **Winning Edge:** GNN + Neo4j

Aligned Govt Schemes: GeM Portal, CVC (Vigilance Commission), Digital India

■ The Problem

On government bidding portals (GeM), vendors form cartels. They coordinate bid prices to inflate costs, taking turns winning at artificially high rates. This drains crores of public taxpayer money. Manual auditing of thousands of tenders is impractical.

■ Proposed Solution

An automated risk-scoring engine monitoring GeM tender/bidding records. It detects collusion by analyzing bidding timelines, price spreads, and hidden corporate connections (shared directors, same registered addresses). Graph Neural Networks classify suspicious vendor sub-graphs that bid together too frequently.

10 Crowdsourced Safety Analytics & CCTV Behavioral Anomaly Monitor

Tags: Public Safety • Women's Safety

Target: Public Safety, Police, Women Welfare | **Feasibility:** 7.0/10 | **Impact:** 9.5/10 | **Winning Edge:** Cross-Camera Re-ID

Aligned Govt Schemes: Nirbhaya Fund, Safe City Project, Smart Cities Mission

■ The Problem

Women's safety in public spaces remains a major concern. CCTV cameras exist but nobody watches feeds 24/7. By the time footage is reviewed, the crime has already happened. An SOS button is useless if the victim can't reach their phone. Real-time automated threat detection is critical.

■ Proposed Solution

An intelligent computer vision layer connecting to existing CCTV feeds. It automatically flags anomalous behavior — a person being stalked across multiple cameras, physical struggles, crowd clustering at odd hours. Sends instant alerts with camera location to nearest police patrols. Automatically blurs non-threat faces for privacy compliance.